

# Information Retrieval 3

## Relevance and Search Evaluation

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Based on: Jimmy Lin, Rodrigo Nogueira, Andrew Yates. *Pretrained Transformers for Text Ranking: BERT and Beyond*.

Synthesis Lectures on Human Language Technologies, Springer, 2021.

**Learning to Rank** = supervised learning based on hand-crafted features. **LambdaMART**, a hybrid of Pointwise/Pairwise/Listwise approaches, was the most successful

**Neural ranking models** = two families: representation-based (DSSM) and interaction-based (DRMM)

**Pre-BERT** neural models struggled to outperform BM25 due to data scarcity

**BERT** = pre-training + fine-tuning achieved approximately 30% relative improvement on MS MARCO

This lecture covers how to define and measure ranking "quality." We systematically study the concepts of **relevance** , **evaluation metrics** , and **test collections**

We learn a methodology for scientifically measuring the "quality" of search results. First, we organize the relationship between text and information needs, then introduce the concept of relevance and evaluation metrics, and finally understand the design and operation of test collections

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**First Half: Text, Information Needs, Relevance**

- Text granularity and corpora
- The gap between information needs and queries
- Formal definition of document ranking
- The concept and subjectivity of relevance
- Relevance judgments (qrels): format and scales

**Second Half: Ranking Metrics and Test Collections**

- Precision, Recall, RR
- Average Precision (AP)
- nDCG (evaluation with graded relevance)
- TREC and the pooling method
- MS MARCO, TREC DL, Robust04

# **Text and Information Needs**

Prerequisite for document ranking: a collection (corpus) of documents  $C = \{d_i\}$

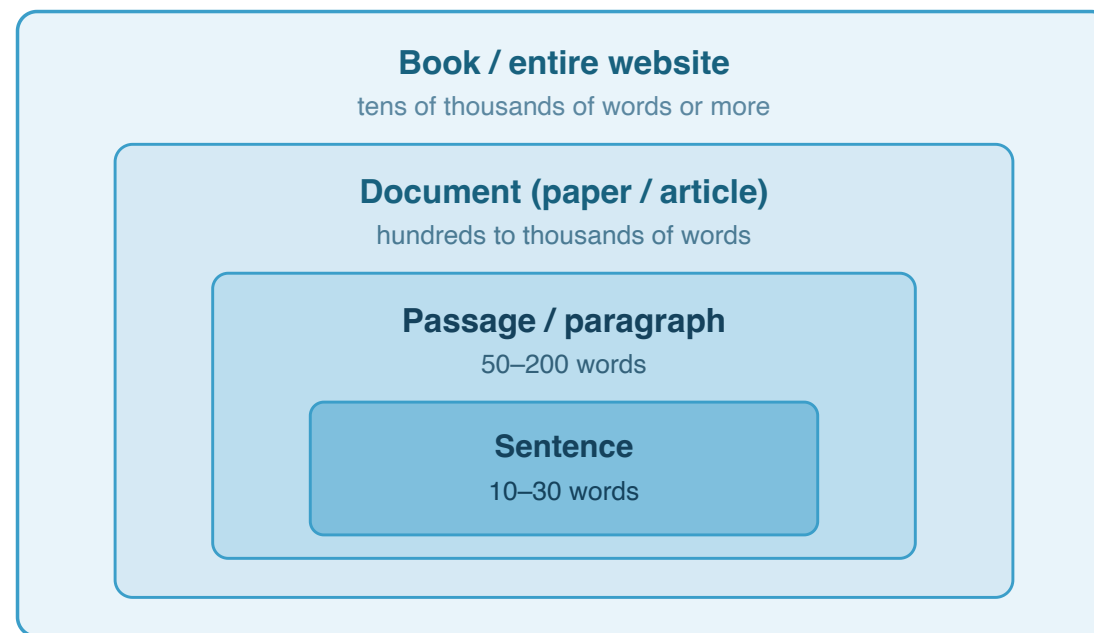
Text length varies widely: from a single CQA sentence to an entire book

The granularity of ranking can be **determined independently**

Example: searching only titles and abstracts in a scientific paper collection

Example: splitting articles into paragraphs and treating each paragraph as a "document"

Transformer input length limits (BERT: max 512 tokens) make the granularity issue even more important



| Corpus            | Documents | Avg. Length | Purpose          |
|-------------------|-----------|-------------|------------------|
| MS MARCO passage  | 8.8M      | 56 words    | Passage ranking  |
| MS MARCO document | 3.2M      | 1,131 words | Document ranking |
| Robust04          | 528K      | 549 words   | Ad hoc retrieval |
| ClueWeb09b        | 50M       | —           | Web search       |
| Common Crawl      | Billions  | —           | Entire web       |

Corpora are provided to the system before queries arrive ( **nearly static** )

At web scale, **computational efficiency** (latency, throughput) considerations are essential

Primarily **unstructured** natural language text (with some structure such as paragraphs and headings, but not tabular data)

The scale of the corpus and the length of documents directly affect ranking model design. Especially for Transformer-based models with input length limits, **document segmentation strategy** becomes an important design decision

The query a user types into a search box is merely an **external manifestation** of an information need

So what is an "information need"? It is more than just "something you want to know"

**Belkin's [1980] ASK Hypothesis:** When a user perceives an **Anomaly** in their knowledge state, they search to resolve it

## **ASK = Anomalous State of Knowledge**

Users do not know precisely "what they don't know" -- that is why they search

An information need begins not as a clear question, but as a **vague cognitive gap**

## Concrete Example of ASK

**Scenario:** Planning a trip to Kyoto next month

**Cognitive gap:**

"I want to see the autumn leaves, but **I don't know when or where peak foliage occurs**"

**Properties of the ASK:**

What I don't know is itself vague

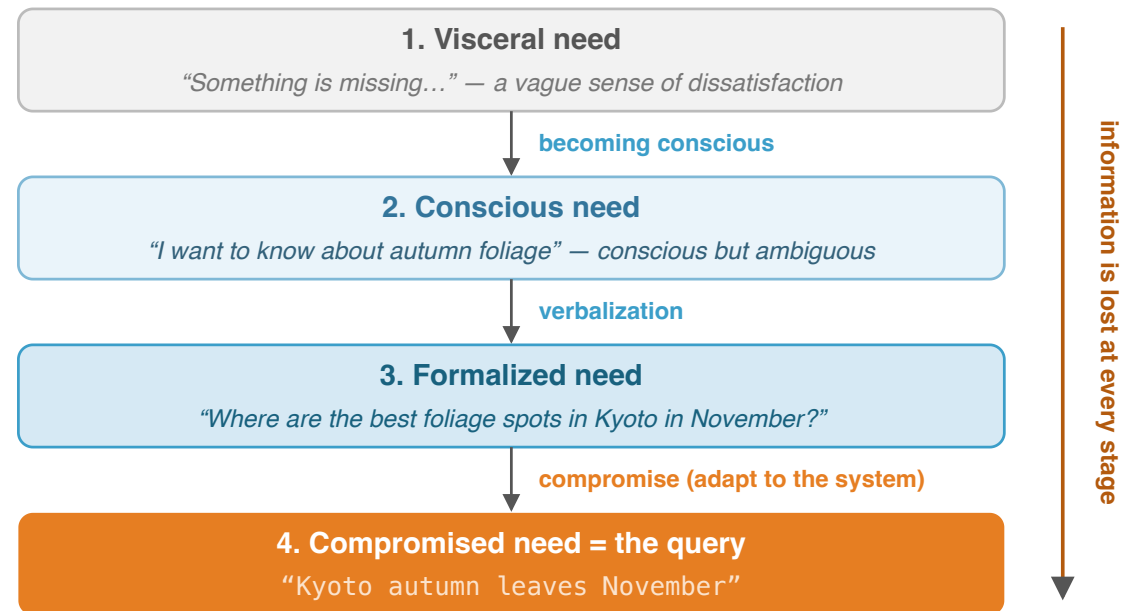
Multiple uncertainties intertwine: "timing," "location," "crowd levels," etc.

The need itself may change during the search process

An ASK (cognitive gap) undergoes **four stages** before being entered into a search system:

1. **Visceral need**: A vague sense of dissatisfaction.  
"Something is missing"
2. **Conscious need**: Conscious but ambiguous. "I want to know about autumn foliage"
3. **Formalized need**: Verbalized. "Places in Kyoto where foliage peaks in November"
4. **Compromised need**: Adapted to the system.  
"Kyoto autumn leaves November"

Information is **lost** at each stage → the query is an **incomplete approximation** of the need





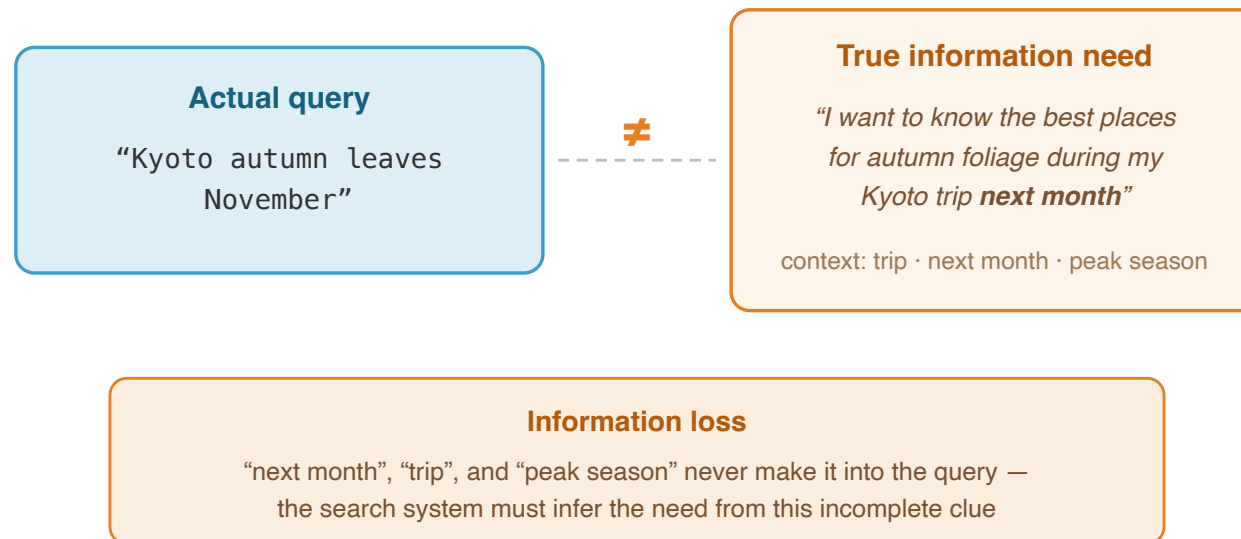
Queries and information needs are **not strictly synonymous**

The same information need can be manifested differently depending on the system:

Web search: "Kyoto autumn leaves November"

Voice assistant: "Tell me the best places for autumn foliage in Kyoto in November"

The form of query expression has a **substantial impact** on ranking effectiveness



**Broder [2002]**'s taxonomy of web search intent:

## Navigational

Want to **go to** a specific site or page  
e.g., "MIT official site," "YouTube"

## Informational ★

Want to **learn about** a specific topic  
e.g., "causes of global warming," "what is BERT"

## Transactional

Want to **do** something (buy, book, download)  
e.g., "book flight tickets," "Python download"

Approximately **80%** of web search queries are informational

IR research primarily targets informational queries (navigational queries are relatively easy since they have a single correct document)

Differences in search intent also affect **the choice of evaluation metrics** (e.g., navigational → MRR, informational → nDCG)

### Document Ranking (Text Ranking)

When an information need is expressed as a query  $q$ , the document ranking task is to return a ranked list of  $k$  documents  $\{d_1, d_2, \dots, d_k\}$  from an arbitrarily large but finite document collection  $C = \{d_i\}$  that maximizes the metric of interest such as nDCG, AP, etc.

#### Components of the definition:

**Input:** Query  $q$  (external manifestation of an information need)

**Target:** Document collection  $C = \{d_i\}$  (corpus)

**Objective:** Maximize the metric of interest (nDCG, AP, etc.)

**Output:** Ranked list of  $k$  documents

$$f : (q, C) \rightarrow [d_1, d_2, \dots, d_k] \quad \text{s.t.} \quad \text{rel}(d_1) \geq \text{rel}(d_2) \geq \dots \geq \text{rel}(d_k)$$

## Ranking

Build a ranked list from the entire corpus  $C$

e.g., BM25 + inverted index

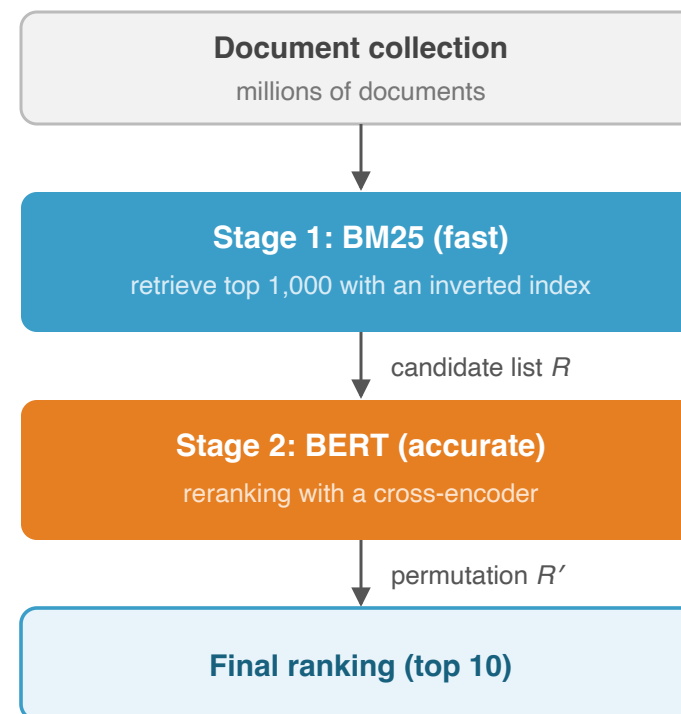
## Reranking

Takes a candidate list  $R$  as input and outputs a permutation  $R'$

e.g., reordering with BERT

Applying Transformers directly to a large-scale corpus is impractical

→ **Multi-stage ranking architecture** is standard



rank cheaply over the whole corpus, then rerank the top accurately

Lin et al., "Pretrained Transformers for Text Ranking: BERT and Beyond", 2021, § 2.2

# Relevance

**Relevance** = a relation between a text and a specific information need

The concept underlying all ranking metrics

Simplest treatment: if a text addresses the information need, it is **relevant** ; otherwise, it is **not relevant**

**Binary definition of relevance:** For a document  $d$  and query  $q$

$$\text{rel}(q, d) = \begin{cases} 1 & \text{(relevant)} \\ 0 & \text{(not relevant)} \end{cases}$$

In practice, it is more complex: it is more accurately characterized on a multi-dimensional ordinal scale

## Example of Relevance Judgments

Query: "machine learning introduction"

✓ **Relevant**

"A Beginner's Guide to Machine Learning"

✓ **Relevant**

"scikit-learn Tutorial"

✗ **Not Relevant**

"Recent Advances in Deep Reinforcement Learning (Advanced)"

IR researchers are primarily concerned with **topical relevance** , but relevance has other important dimensions

## Topical Relevance

Whether the topic of the text **matches** the information need  
Primary focus of IR research

## Cognitive Relevance

Whether the text is **comprehensible** to the user  
e.g., a research paper is not relevant to a beginner

## Situational Relevance

Whether the text is **useful** for problem-solving  
e.g., a closed-restaurant listing is situationally irrelevant

## Concrete examples:

Searching for "quantum computing" → A cutting-edge research paper is **topically relevant** but **cognitively irrelevant** to a high school student

Searching for "Tokyo restaurants" → An article about a closed restaurant is **topically relevant** but **situationally irrelevant**

Saracevic, "Relevance: A Review of the Literature and a Framework for Thinking on the Notion in Information Science", 2017

The most important lesson about relevance:

## Relevance is in the eye of the beholder

Relevance judgments are user-specific subjective judgments, neither a Platonic "truth" nor an "inherent property" of a document

People can have different concepts of relevance, and all are "correct"

This is qualitatively different from NLP manual annotation (e.g., POS tags)

### POS tagging (NLP annotation)

"run" → verb

an objectively correct answer exists

**high agreement ( $\kappa > 0.9$ )**

### Relevance judgment (IR)

"Is this document useful?" → it depends

inherently subjective, user-specific

**low agreement (overlap  $\approx 60\%$ )**



Inter-annotator agreement is **surprisingly low**

Overlap rate  $\approx 60\%$  is a frequently cited figure

Actual range: 30% to over 70% (depends on study design, information need, and agreement metric used)

Much lower than NLP manual annotation ( $\kappa > 0.9$  is the benchmark for "good" agreement in POS tagging, etc.)

Note that 60% is a simple overlap rate, not a  $\kappa$  coefficient that corrects for chance agreement

## Reasons for low agreement:

Topic descriptions themselves are incomplete:  
information needs are inherently cognitive states (ASK) in the user's mind

**Relevance paradox:** if a user could fully verbalize their information need, there would be no need to search

## Key Finding: Stability of System Rankings

The **absolute values** of relevance judgment scores vary across judgment sets, but the **relative ordering among systems** is stable

Even with different assessors' judgment sets,  
**Kendall's  $\tau > 0.9$**  is consistently observed  
→ The conclusion "System A is better than B"  
holds even when the assessors change

Despite low inter-annotator agreement, IR evaluation holds as a science for the following reasons:

1. **Stability of relative comparisons:** The superiority relationship among systems is consistent across different judgment sets (Kendall's  $\tau > 0.9$ )
2. **Averaging over many topics:** Noise on individual topics is canceled out by averaging over many topics
3. **Diversity of pooling:** Including runs from diverse methods reduces bias

## Stability of System Rankings

Assessor A's judgments:

**Sys1 > Sys2 > Sys3 > Sys4**

Assessor B's judgments:

**Sys1 > Sys2 > Sys3 > Sys4**

Absolute scores differ, but

**the ordering is consistent**

**Relevance judgments do not need to be "perfect." It suffices that relative comparisons among systems are reliable**

# Relevance Judgments

Relevance Judgments (qrels)

**qrels** = a set of triples  $(q, d, r)$

$q$ : query (topic)

$d$ : document

$r$ : relevance judgment (human annotation)

Downloadable as part of a test collection

Also known as: relevance labels, human judgments

### Example in TREC format:

```
336 0 FBIS3-10082 1
336 0 FBIS3-10332 0
336 0 LA092790-0015 2
```

topic\_id iteration doc\_id relevance

### Structure of qrels

| Field            | Description                    |
|------------------|--------------------------------|
| <b>topic_id</b>  | Query identifier               |
| <b>iteration</b> | Usually 0 (for compatibility)  |
| <b>doc_id</b>    | Document identifier            |
| <b>relevance</b> | Relevance grade (0, 1, 2, ...) |

The evaluation program (**trec\_eval**) takes qrels and system output (run) as input and computes various metrics

## Binary

Relevant or Not Relevant

Simplest. Used with Precision, AP, etc.

## Three-level

Not Relevant / Relevant / Highly Relevant

Simplified evaluation in TREC DL

## Multi-level (PEGFB, etc.)

Perfect / Excellent / Good / Fair / Bad

Used for nDCG evaluation

Non-binary judgments are called **graded relevance judgments**

TREC Deep Learning Track uses **4-level** relevance judgments

NTCIR (an evaluation event in Japan and Asia) also uses graded judgments

e.g., NTCIR-9 INTENT-1 task: L0 (not relevant), L1 (somewhat relevant), L2 (relevant), L3 (highly relevant)

Some metrics can leverage graded relevance (nDCG), while others cannot (Precision, AP). This difference must be considered when choosing metrics

## Use 1: Model Training

Used as labels for training ranking models with supervised learning

### Training data format:

- (query, relevant doc) pairs
- (query, relevant doc, non-relevant doc) triples

## Use 2: Model Evaluation

Used as a gold standard for quantitatively measuring ranking model quality

### Evaluation workflow:

- Compare system output (run) against qrels
- Compute various metrics using trec\_eval, etc.

Historically, IR test collections were not large enough for training (only for parameter tuning)

The release of **MS MARCO** (2016) enabled large-scale supervised learning for the community

**Negative sampling problem:** Absence of a relevance label  $\neq$  confirmation of non-relevance. In MS MARCO, non-relevant documents are sampled from BM25 results (pseudo-labels). The sampling method significantly affects model quality

# Ranking Metrics

Ranking Metrics

Ranking metrics = functions that **quantify** the quality of a ranked list from relevance judgments (qrels)

Notation:

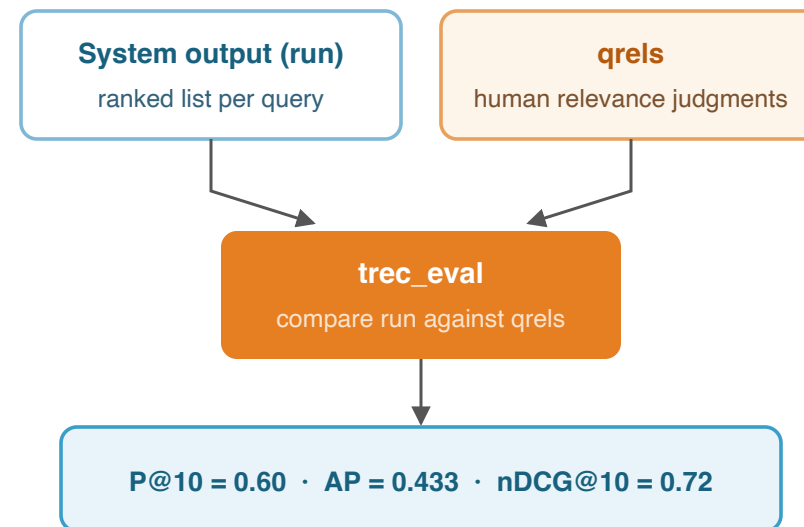
Ranked list  $R = \{(i, d_i)\}_{i=1}^l$

$\text{rel}(q, d)$ : relevance of document  $d$  to query  $q$

Truncated at cutoff  $k$ :  $\text{Metric}@k$

## Desirable properties of a metric:

1. Higher score for lists containing more relevant documents
2. Higher score for lists with relevant documents ranked higher
3. Ability to distinguish graded relevance is desirable



a metric maps (run, qrels) to a single quality score per query



$$\text{Precision}(R, q) = \frac{\sum_{(i,d) \in R} \text{rel}(q, d)}{|R|}$$

The **proportion of relevant documents** in the ranked list  $R$

$\text{rel}(q, d)$ : binary (relevant=1 / not relevant=0)

**Precision@  $k$  (P@  $k$  )**: precision at the top  $k$  results

**Drawbacks:**

1. Cannot leverage graded relevance judgments
2. Does not consider rank positions beyond the cutoff

**P@5 Calculation Example**

| Rank | Document | Relevant? |
|------|----------|-----------|
| 1    | $d_1$    | ○         |
| 2    | $d_2$    | ○         |
| 3    | $d_3$    | ×         |
| 4    | $d_4$    | ○         |
| 5    | $d_5$    | ×         |

$$\text{P@5} = \frac{3}{5} = 0.60$$

3 out of the top 5 are relevant → 60% precision

$$\text{Recall}(R, q) = \frac{\sum_{(i,d) \in R} \text{rel}(q, d)}{\sum_{d \in C} \text{rel}(q, d)}$$

The proportion of relevant documents in the entire collection that are **included in** the list  $R$

Denominator: total number of relevant documents in corpus  $C$

**Recall@  $k$  ( $R@ k$  )**: recall at the top  $k$  results

### Relationship between Precision and Recall:

Precision: proportion of relevant documents among returned results ( **accuracy** )

Recall: proportion of relevant documents not missed ( **coverage** )

### R@5 Calculation Example

Total relevant documents in corpus = **8**

Relevant documents in top 5 = **3**

$$R@5 = \frac{3}{8} = 0.375$$

Found 3 out of 8 relevant documents in the top 5 → 37.5% recall

The remaining 5 relevant documents are ranked below position 6 or not in the list

As the cutoff  $k$  increases:

**Recall tends to increase** (may include more relevant documents)

**Precision tends to decrease** (non-relevant documents also increase)

This is called the **Precision-Recall tradeoff**

**R-precision:** Precision at cutoff  $k$  set to the number of relevant documents for that topic  $|\text{Rel}|$

$$\text{R-precision} = P@|\text{Rel}|$$

Corresponds to the point where Precision equals Recall

Changes in P/R with Cutoff

| $k$ | Rel. Count | $P@k$ | $R@k$ |
|-----|------------|-------|-------|
| 1   | 1          | 1.00  | 0.13  |
| 3   | 2          | 0.67  | 0.25  |
| 5   | 3          | 0.60  | 0.38  |
| 10  | 5          | 0.50  | 0.63  |
| 20  | 6          | 0.30  | 0.75  |

Total relevant documents = 8

$$RR(R, q) = \frac{1}{\text{rank}_1}$$

$\text{rank}_1$  = **rank position of the first relevant document**

Intuitive and easy to compute

| Position of First Relevant Doc | RR    |
|--------------------------------|-------|
| 1st                            | 1.000 |
| 2nd                            | 0.500 |
| 3rd                            | 0.333 |
| 5th                            | 0.200 |
| No relevant doc                | 0.000 |

### Suitable scenarios:

QA (Question Answering): tasks where one answer suffices

Navigational search: tasks looking for a specific page

### Limitations:

Captures **only** the first relevant result

Completely ignores the positions of relevant documents from rank 2 onward

**Inappropriate** for ad hoc search (which requires multiple relevant documents)

**MRR (Mean Reciprocal Rank)** = the arithmetic mean of RR across all topics. The official evaluation metric for MS MARCO passage is **MRR@10**

$$AP(R, q) = \frac{\sum_{(i,d) \in R} P@i \cdot \text{rel}(q, d)}{\sum_{d \in C} \text{rel}(q, d)}$$

### Intuitive understanding:

Compute Precision at each position where a **relevant document appears**, then take their average

$\text{rel}(q, d)$  is a binary indicator  $\rightarrow$  Precision is not added at positions of non-relevant documents

The denominator is the total number of relevant documents in the corpus  $\rightarrow$  Relevant documents not in the list **contribute 0**

### Advantages of AP:

A summary statistic that captures **both Precision and Recall aspects** in a single number

Considers rank position (higher score when relevant documents are concentrated at the top)

### What AP Captures

**Relevant documents concentrated at the top**  $\rightarrow$  High AP

Because  $P@i$  at each position is large

**Relevant documents scattered**  $\rightarrow$  Low AP

Because intervening non-relevant documents lower  $P@i$

**Drawback of AP:** Cannot distinguish graded relevance. "Somewhat relevant" and "highly relevant" contribute equally

**Setup:** Total relevant = 5 , 3 appear in the top-10 list

| Rank $i$ | Rel? | $P@i$         | Contribution |
|----------|------|---------------|--------------|
| 1        | ○    | $1/1 = 1.000$ | 1.000        |
| 2        | ×    | —             | —            |
| 3        | ○    | $2/3 = 0.667$ | 0.667        |
| 4-5      | ×    | —             | —            |
| 6        | ○    | $3/6 = 0.500$ | 0.500        |
| 7-10     | ×    | —             | —            |

$$AP = \frac{1.000 + 0.667 + 0.500}{5} = \frac{2.167}{5} = 0.433$$

Denominator = 5 (total relevant). The 2 unjudged relevant docs contribute  $P = 0$

**MAP** = arithmetic mean of AP across all topics

The most common metric that leverages **graded relevance**

$$\text{DCG}(R, q) = \sum_{(i,d) \in R} \frac{2^{\text{rel}(q,d)} - 1}{\log_2(i + 1)}$$

**Two components:**

**1. Gain:**  $2^{\text{rel}} - 1$

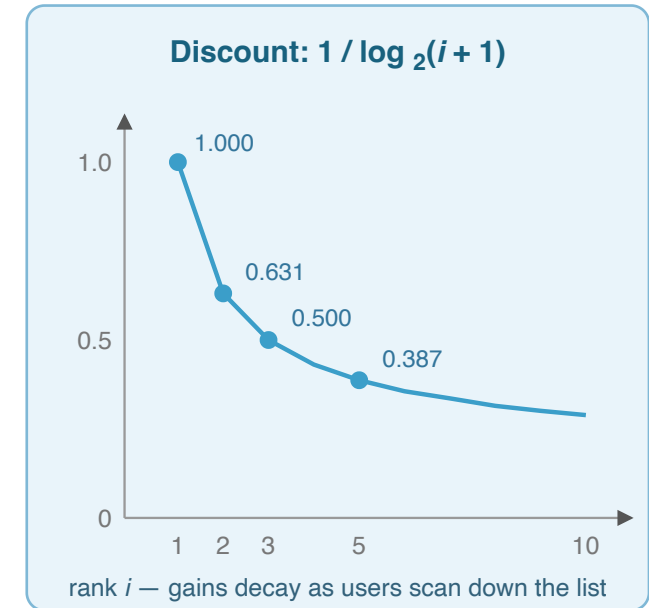
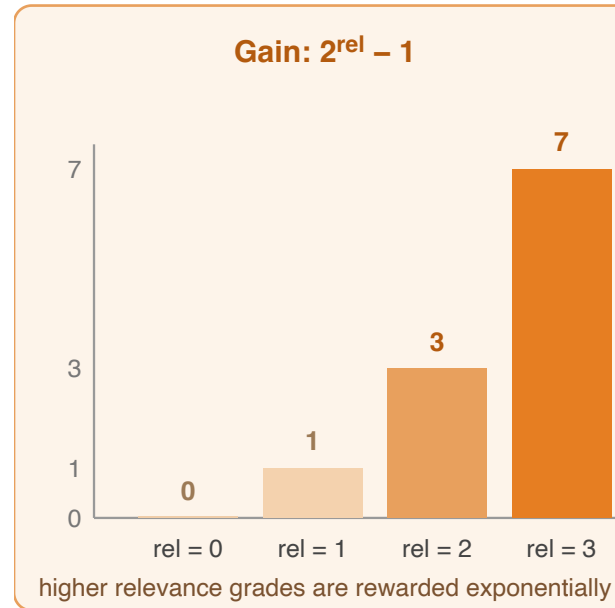
Higher value for higher relevance grades

rel=0 → 0, rel=1 → 1, rel=2 → 3, rel=3 → 7

**2. Discount:**  $\frac{1}{\log_2(i+1)}$

Attenuates gain as rank decreases

Users pay more attention to top results



DCG values depend on the distribution of relevant documents per query, making cross-query comparisons impossible → **normalization** is needed

$$\text{nDCG}(R, q) = \frac{\text{DCG}(R, q)}{\text{IDCG}(R, q)}$$

**IDCG (Ideal DCG)** = DCG of the **ideal ranking**

where documents with the highest relevance are placed first

nDCG is normalized to **[0, 1]**

1.0 = perfect ranking

0.0 = no relevant documents

**Commonly used cutoffs:**

| Cutoff  | Use Case                    |
|---------|-----------------------------|
| nDCG@3  | Mobile search               |
| nDCG@5  | Mobile search               |
| nDCG@10 | Web search (most common)    |
| nDCG@20 | Two pages of search results |

**Rationale for discount:** Users scan search results from top to bottom, with increasing probability of "giving up" at lower ranks [Joachims et al., 2007]



**Setup:** Query "information retrieval textbook", graded relevance  $\text{rel} \in \{0, 1, 2, 3\}$

Actual Ranking → DCG

| $i$ | rel | Gain | Discount | Component |
|-----|-----|------|----------|-----------|
| 1   | 3   | 7    | 1.00     | 7.00      |
| 2   | 0   | 0    | 1.58     | 0.00      |
| 3   | 2   | 3    | 2.00     | 1.50      |
| 4   | 1   | 1    | 2.32     | 0.43      |
| 5   | 2   | 3    | 2.58     | 1.16      |

DCG@5 = 10.09

Ideal Ranking → IDCG

| $i$ | rel | Gain | Discount | Component |
|-----|-----|------|----------|-----------|
| 1   | 3   | 7    | 1.00     | 7.00      |
| 2   | 2   | 3    | 1.58     | 1.90      |
| 3   | 2   | 3    | 2.00     | 1.50      |
| 4   | 1   | 1    | 2.32     | 0.43      |
| 5   | 0   | 0    | 2.58     | 0.00      |

IDCG@5 = 10.83

$$\text{nDCG@5} = \frac{\text{DCG@5}}{\text{IDCG@5}} = \frac{10.09}{10.83} \approx 0.932$$

A non-relevant document at rank 2 causes about a 7% drop from the ideal. However, the correct placement of the relevance-3 document at rank 1 is rewarded

| Metric   | Relevance Type | Rank Position Consideration | Normalization | Primary Use                       |
|----------|----------------|-----------------------------|---------------|-----------------------------------|
| $P@k$    | Binary         | None                        | None          | Simple top-k precision evaluation |
| $R@k$    | Binary         | None                        | None          | Coverage evaluation               |
| RR       | Binary         | First relevant only         | None          | QA, navigational search           |
| AP       | Binary         | Yes (all relevant docs)     | Recall-like   | Ad hoc search (TREC standard)     |
| $nDCG@k$ | Graded         | Yes (discounted)            | IDCG          | Web search (most common)          |

**Metric selection guide:** Binary relevance judgments → AP. Graded relevance judgments → nDCG. Tasks seeking a single answer → RR. These metrics are computed per query, and the **average** across all queries is reported (MAP, MRR, mean nDCG)

## Statistical Testing:

Need to verify whether differences between systems are **statistically significant**

Main testing methods:

- Paired t-test

- Randomization test

- Bootstrap test

Per-topic scores are treated as paired data

## Score Ties:

All metrics assume **strictly monotonically decreasing** scores with no ties

The impact of ties is small (around the 3rd decimal place), but systems should ensure no ties occur

## Unjudged Documents:

Documents not in qrels = **unjudged documents**

Standard treatment: **assumed not relevant**

**Problem:** If a ranking model discovers a novel relevant document, but that document is unjudged, it will not be reflected in the score

**Diagnostic method:** Compute  $\text{Judedged@}k$  ( $J@k$ ) to check the proportion of judged documents in the top  $k$

This problem is particularly severe for post-hoc runs that did not participate in pooling (discussed below)

# Test Collections

Test Collections

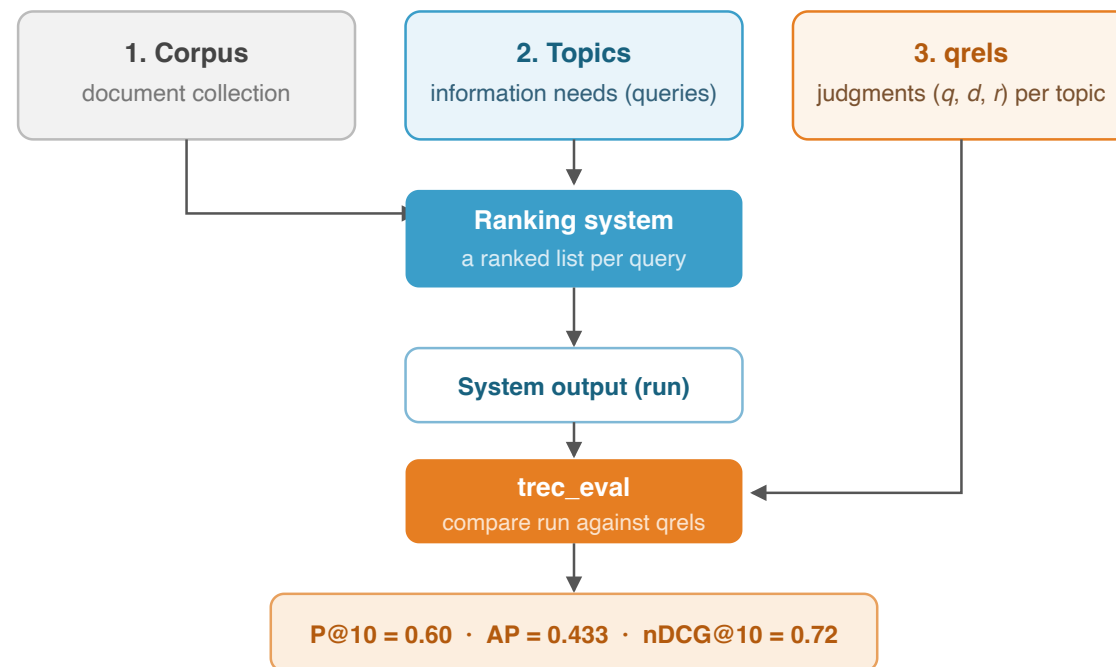
The **three components** needed for evaluating document ranking models:

## Test Collection

1. **Corpus**: the document collection to be searched
2. **Topics**: the set of information needs
3. **Relevance judgments (qrels)**: relevance judgments of documents for each topic

With a test collection, ranking model quality can be quantitatively measured

Known as the **Cranfield paradigm** (established by Cleverdon in the 1960s)



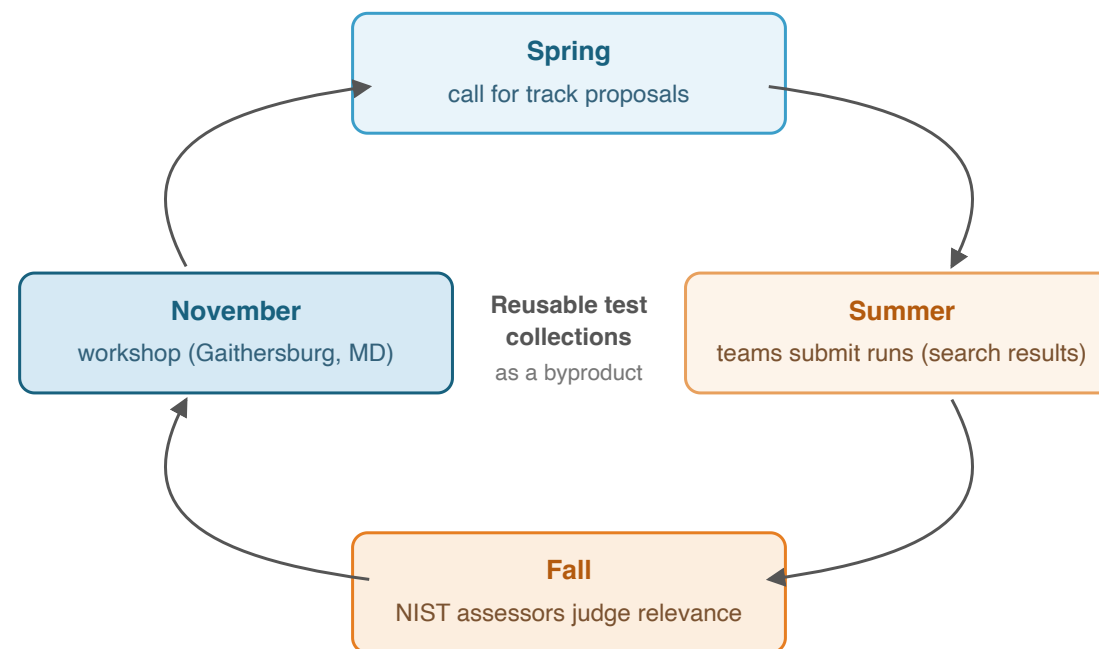
**TREC (Text REtrieval Conference):** organized by the National Institute of Standards and Technology (NIST) in the U.S.

**Running continuously for over 30 years since 1992**, the world's largest IR evaluation event

Participation is completely voluntary (no prizes or monetary incentives)

### Purposes of TREC:

1. An opportunity for the community to collectively set the agenda
2. A forum for fair and neutral system evaluation
3. **Reusable test collections** as a byproduct
4. A testbed for advancing evaluation methodology itself



## TREC's real-world impact:

Google's early days (early 2000s): algorithms based on TREC research were adopted in web search

IBM Watson (2011): defeated humans on Jeopardy! -- a direct descendant of the TREC QA track

## Global spread of the TREC model:

**CLEF:** European IR evaluation forum

**NTCIR:** IR evaluation for Japan and Asia (organized by NII)

IR for Asian languages including Japanese, Chinese, Korean

Running since 1999

**FIRE:** Indian IR evaluation forum

## NTCIR -- IR Evaluation in Japan

**Organizer:** National Institute of Informatics (NII)

**Started:** 1999 (NTCIR-1)

### Major tasks:

CLIR (Cross-Language IR)

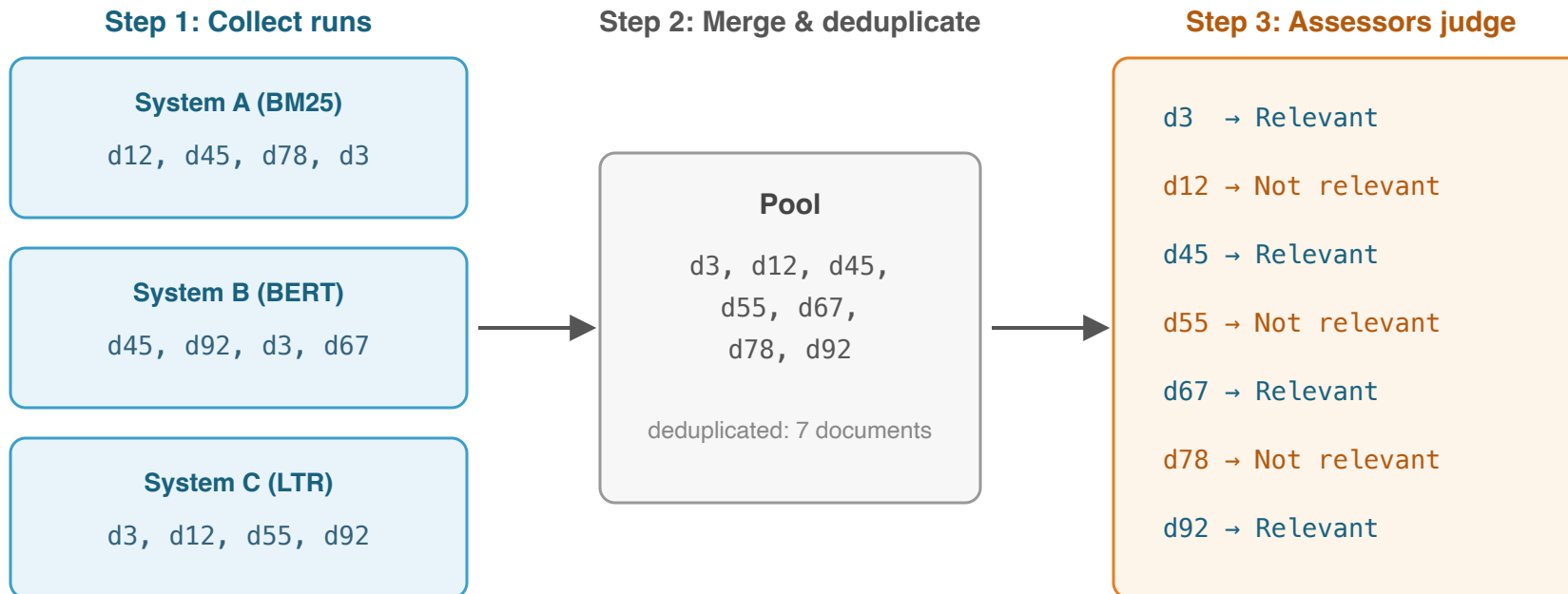
Patent retrieval

QA Lab (university entrance exam QA)

WWW task (Yahoo! Japan data)

Important contributions to statistical testing methods for IR evaluation metrics have also emerged from this community

Assigning relevance judgments to all documents in the entire corpus is **impossible** → **Pooling** efficiently samples documents for judgment



**Pool diversity matters:** runs from different methods produce qrels that are not biased toward one algorithm

**Post-hoc run problem:** novel documents from systems outside the pool remain unjudged → risk of unfairly low scores

Sparck Jones and van Rijsbergen, "Report on the Need for and Provision of an Ideal Information Retrieval Test Collection", 1975; Zobel, "How Reliable Are the Results of Large-Scale Information Retrieval Experiments?", SIGIR, 1998



Released by **Microsoft** in 2016

**The pivotal resource that triggered the BERT-driven document ranking revolution**

Corpus: **8.8 million passages** extracted from Bing search results

Queries: extracted from the Bing query log (primarily informational)

**Characteristics of judgments -- shallow but wide:**

On average **1** relevant passage per query

Training set: **500K queries** × 1 relevant passage

Official evaluation metric: **MRR@10**

### MS MARCO Statistics

| Item             | Value      |
|------------------|------------|
| Documents        | 8.8M       |
| Avg. doc length  | 56.3 words |
| Training queries | 502,939    |
| Relevant docs/q  | 1.06       |

Real user queries → a more "natural" distribution of information needs than artificial datasets such as SQuAD

## Why MS MARCO matters:

Provided large-scale **supervised learning data** for the academic community

Previously, data at this scale was available only within search engine companies (Google, Bing, etc.)

Became the **standard training data** for many Transformer-based ranking models

**Zero-shot transfer capability** to out-of-domain tasks has also been confirmed

## Sparse judgment problem:

1. Negative examples are sampled from BM25 results, assumed to be non-relevant → **pseudo-labels**
2. One relevant document per query is too coarse → may fail to properly evaluate the effect of query expansion
3. Absence of a relevant label  $\neq$  confirmation of non-relevance

**The TREC Deep Learning Track was designed to complement the sparse judgments of MS MARCO**

**Started in 2019:** the first large-scale community comparison of pre-BERT vs post-BERT systems  
Uses the MS MARCO corpus (passage + document tasks)

#### Differences from MS MARCO:

**4-level graded relevance judgments** (deep judgments via pooling)

On average **over 200** judgments per query

Official metrics: **nDCG@10, MAP**

Fewer queries (43-54) but dense judgments

#### TREC DL Statistics

| Item             | 2019    | 2020    |
|------------------|---------|---------|
| Queries          | 43      | 54      |
| Avg. judgments/q | 215     | 211     |
| Relevant docs/q  | 58.2    | 30.9    |
| Relevance scale  | 4-level | 4-level |

In contrast to MS MARCO's shallow but wide (many queries, few judgments), TREC DL is narrow but deep (few queries, many judgments)

| Aspect              | MS MARCO         | TREC Deep Learning |
|---------------------|------------------|--------------------|
| Queries             | 500K+            | 43-54              |
| Judgments/query     | ~1 (sparse)      | 200+ (dense)       |
| Relevance scale     | Binary           | 4-level            |
| Primary use         | <b>Training</b>  | <b>Evaluation</b>  |
| Official metric     | MRR@10           | nDCG@10, MAP       |
| Nature of judgments | shallow but wide | narrow but deep    |

**Practical workflow:** **Train models** on MS MARCO's large-scale data, and **evaluate models** with TREC DL's deep judgments. This combination became the standard experimental setup for IR research in the BERT era

One of the **most widely used ad hoc retrieval test collections** (~20 years of use)

Corpus: **newswire** (TREC Disks 4 & 5) — approximately **528,000 documents**

**249 topics** , approximately **311,000** relevance judgments (69.9 rel/query)

### Strengths:

Diverse pools (including manual runs) → **high reusability**

Used in **over 100 papers** by early 2019

Can be used for training with 5-fold cross-validation

### Robust04 Statistics

| Item              | Value              |
|-------------------|--------------------|
| Documents         | 528,155            |
| Avg. doc length   | 548.6 words        |
| Topics            | 249                |
| Avg. query length | 2.67 words (title) |
| Judgments         | 311,410            |
| Relevant docs/q   | 69.9               |

Approximately 20 years of history → high **risk of overfitting**. No standard train/test split has been established

## Limitations of the Cranfield paradigm:

Reduces the ranking problem to optimization of quantitative metrics

**Ecological validity problem:** do the information needs in test collections reflect real-world user needs?

Improved search accuracy  $\neq$  user task success

e.g., improved medical literature search accuracy does not necessarily improve clinician diagnostic accuracy

## Dangers of overfitting:

**p-hacking:** trying many model variants and reporting only the best results

**File drawer problem:** experiments that did not work are not reported

Repeated use of test collections  $\rightarrow$  implicit fitting to the benchmark

## Countermeasures:

TREC participation: test sets are used only once (blind)

Private test sets on leaderboards (MS MARCO)

Demonstrate **consistent improvement across multiple test collections**

1. **Text and information needs:** A query is an incomplete manifestation of an information need. Understanding text granularity and search intent classification is the foundation for ranking model design
2. **Relevance:** Has topical, cognitive, and situational dimensions and is inherently subjective. However, relative comparisons between systems are stable (Kendall's  $\tau > 0.9$ )
3. **Ranking metrics:** AP for binary relevance, nDCG for graded relevance. The choice of metric depends on the task and relevance scale
4. **Test collections:** Three components: corpus + topics + qrels. Training on MS MARCO and evaluating on TREC DL is the standard in the BERT era

## Preview: Information Retrieval 4 -- Traditional Retrieval Models

Boolean model

Vector Space Model (VSM) and TF-IDF

Probabilistic model (BIM) and language model